1. Differentiation

1.1. Basic Derivatives

x^n	nx^{n-1}
e^u	$\frac{\mathrm{du}}{\mathrm{dx}}e^u$
$\ln u$	$\frac{\frac{\mathrm{du}}{\mathrm{dx}}}{u}$
$\sin(ax)$	$a\cos(ax)$
$\cos(ax)$	$-a\sin(ax)$
$\tan(ax)$	$a\sec^2(ax)$
$tan^{-1}(ax)$	$\frac{a}{1+(ax)^2}$

1.2. Chain, Product and Quotient Rule

- Chain Rule:

$$\frac{\mathrm{dy}}{\mathrm{dx}} = \frac{\mathrm{dy}}{\mathrm{du}} \times \frac{\mathrm{du}}{\mathrm{dx}}$$

- Product Rule:

$$\frac{d}{\mathrm{dx}}(\mathrm{uv}) = u\frac{\mathrm{dv}}{\mathrm{dx}} + v\frac{\mathrm{du}}{\mathrm{dx}}$$

- Quotient Rule:

$$\frac{d}{\mathrm{dx}}\left(\frac{u}{v}\right) = \frac{v\frac{\mathrm{du}}{\mathrm{dx}} - u\frac{\mathrm{dv}}{\mathrm{dx}}}{v^2}$$

1.3. Parametric Equations

$$\frac{\mathrm{dy}}{\mathrm{dx}} = \frac{\mathrm{dy}}{\mathrm{dt}} \div \frac{\mathrm{dx}}{\mathrm{dt}}$$

- In a parametric equation x and y are given in terms of t and you must use the above rule to find the derivative

1.4. Implicit Functions

- These represent circles or lines with circular curves, on a Cartesian plane
- Difficult to rearrange in form $y = \therefore$ differentiate as is
- Differentiate x terms as usual
- For y terms, differentiate the same as you would x but multiply with $\frac{\mathrm{dy}}{\mathrm{dx}}$
- Then make $\frac{\mathrm{dy}}{\mathrm{dx}}$ the subject of formula for derivative

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